welcome

Artificial Intelligence
CSE - 311

Genetic Algorithm

Contents

Evolution?

- Evolution: A series of genetic changes by which a living organism acquires characteristics that distinguish it from other organisms.

- Human are the products of evolution.

- So, by modeling the process of evolution, we might expect to create intelligent behaviour.

Natural Evolution

- On 1st July 1858, Darwin Presented his theory of evolution
- Darwin's classical theory of evolution, together with Weisman's theory of natural selection and Mendel's concept of genetics now represent the Neo-Darwinian paradigm.
- Neo-Darwinism is based on the process of reproduction, mutation, competition and selection.

Evolutionary Computation

- Evolutionary computation simulates evolution on a computer.
- The evolution approach to machine learning is based on computational model of natural selection and genetics. We call them evolutionary computation, an umbrella term that combines genetic algorithms, evolution strategies and genetic programming.

Genetic Algorithms

- A type of evolutionary computation inspired by Darwin’s theory of evolution.
- A genetic algorithm generates a population of possible solutions encoded as chromosomes, evaluates their fitness, and create a new population by applying genetic operators-crossover and mutation.
- By repeating this process over many generations, the genetic algorithm breeds an optimal solution to the problem.

Keywords Related to GA

- **Gene:** A basic unit of chromosome that controls the development of a particular feature of a living organism. In Holland’s (Founder of GA) Chromosome a gene is represented by either 0 or 1.
- **Chromosomes:** A string of genes that represent an individual. Each chromosome consists of a number of gene.
- **Fitness:** The ability of a living organism to survive and reproduce in a specific environment.
- **Crossover:** A reproduction operator that creates a new chromosome by exchanging parts of two existing chromosome.
- **Mutation:** A genetic operator that randomly change the gene value in a chromosome.

Keywords Related to GA (cont.)

- **Fitness Function**: A mathematical function used for calculating the fitness of a chromosome.
- **Crossover Probability**: A number between zero and one that indicates the probability of two chromosomes crossing over.
- **Mutation Probability**: A number between zero and one that indicates the probability of mutation occurring in a single gene.
- **Offspring**: An individual that was produced through reproduction. It also referred to as child.
- **Population**: A group of individuals that breed together.

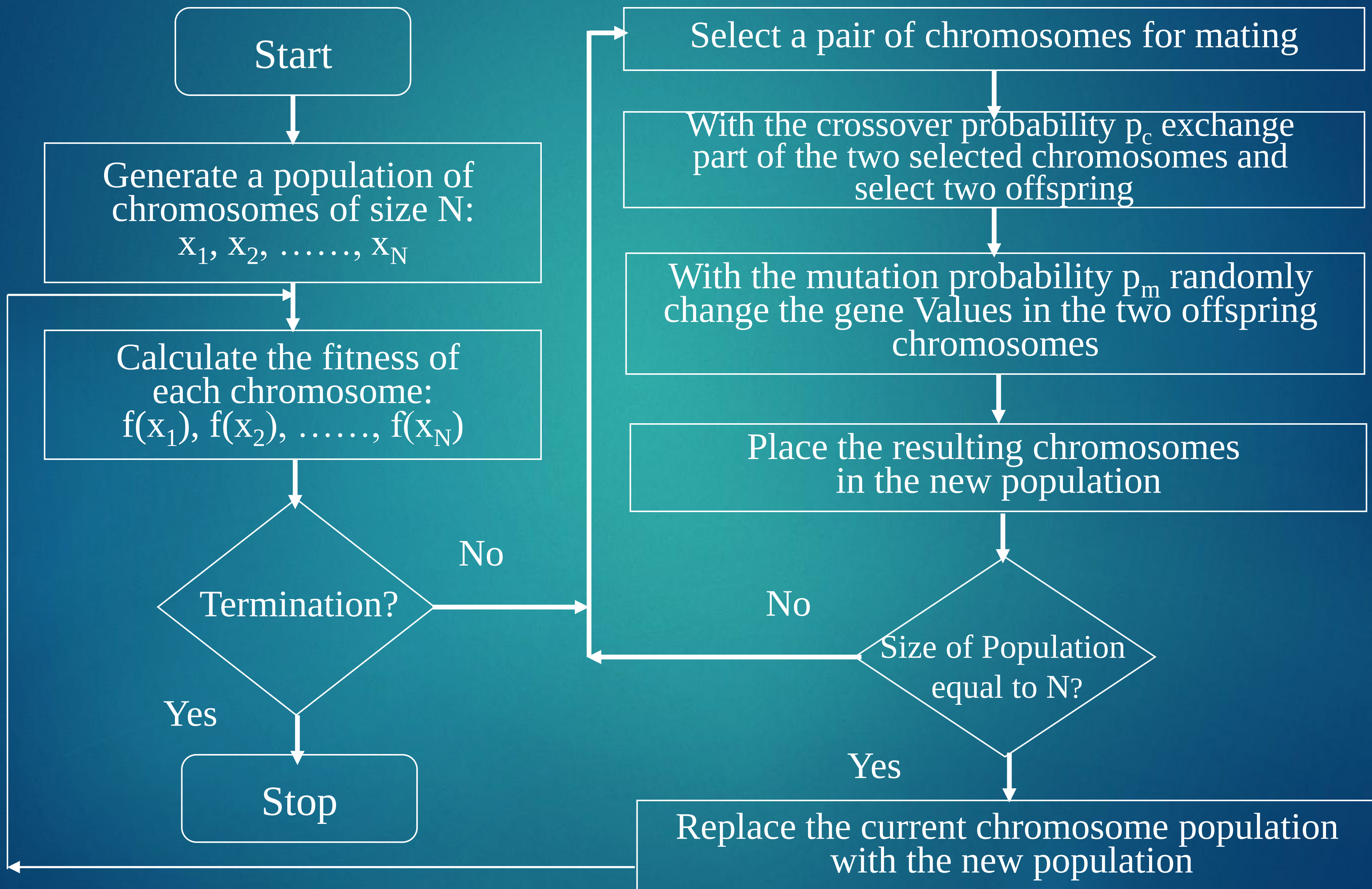
Evolution Strategy

- Evolution strategy is a numerical optimization procedure.
- Unlike the genetic algorithms, evolution strategies use only a mutation operator, and do not require a problem to be represented in coded form.
- Evolution strategies are used for solving technical optimization problems when no analytical objective function is available, and no conventional optimization method exists.

Genetic Programming

- An application of genetic algorithms to computer programs.
- Genetic programming is most easily implemented where the programming language permits a program to be manipulated as data and the newly created data to be executed as a program.

Genetic Algorithm Flowchart



Example Problem & Solution

- Find the maximum value of the function ($15x-x^2$), where parameter x varies between 0 and 15.
  - Chromosome can be built with only four gene (0-15)

Solution:

Suppose,

- The size of the chromosome population, N = 6
- The crossover probability, pc = 0.7
- The mutation probability, pm = 0.001
- The fitness function, $f(x) = 15x-x^2$

Example Problem & Solution

Step 1: GA creates an initial population of chromosomes by filling six 4-bit string with randomly generated ones and zeros

Chromosome Label	Chromosome String	Decoded Integer	Chromosome Fitness	Fitness Ratio
X1	1100	12		
X2	0100	4		
X3	0001	1		
X4	1110	14		
X5	0111	7		
X6	1001	9		

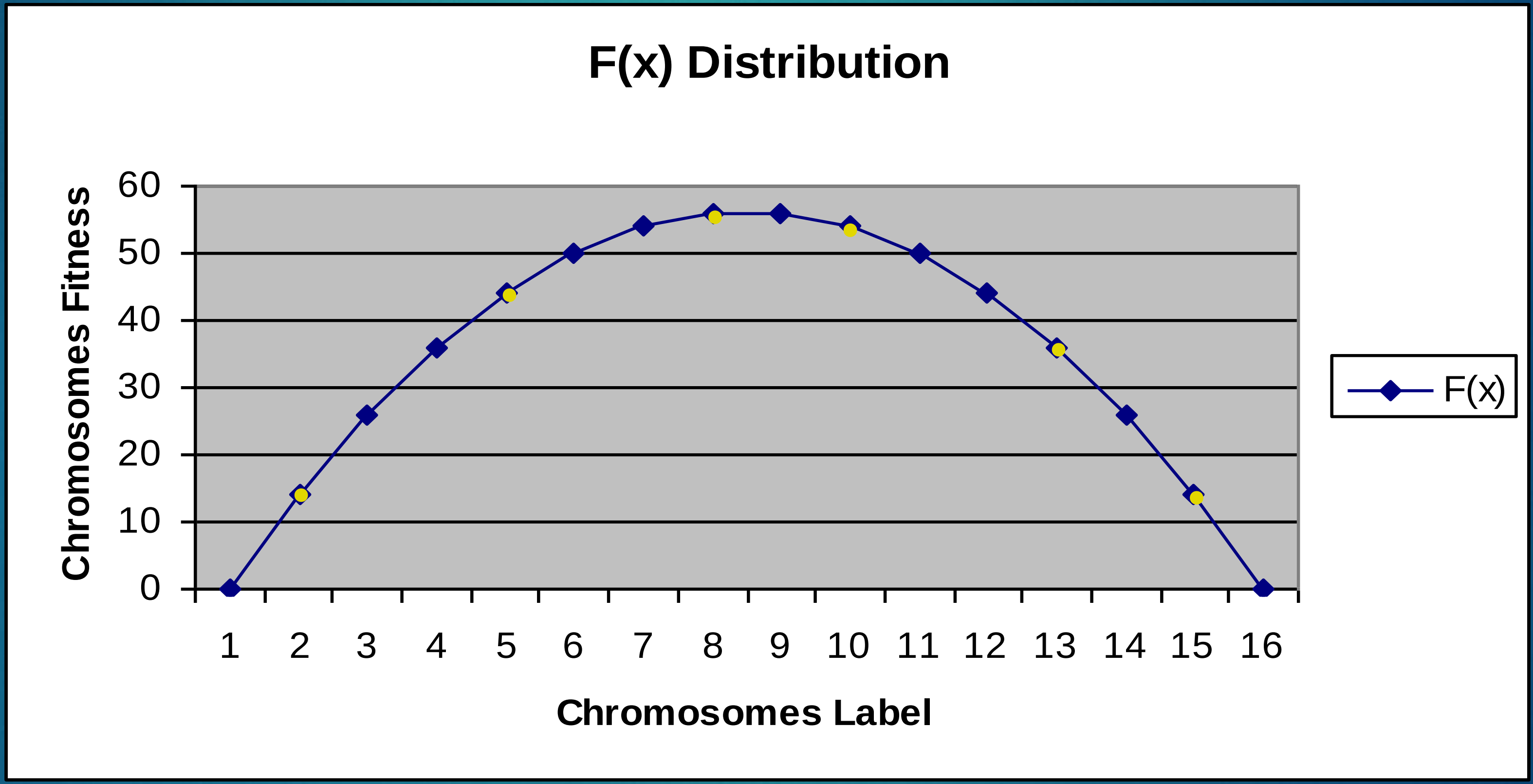
Example Problem & Solution

Step 2: Calculate the fitness of each individual Chromosome

Chromosome Label	Chromosome String	Decoded Integer	Chromosome Fitness	Fitness Ratio
X1	1100	12	36	
X2	0100	4	44	
X3	0001	1	14	
X4	1110	14	14	
X5	0111	7	56	
X6	1001	9	54	

$$15x-x^2 = 15*12 - 12^2 = 180 - 144 = 36$$

Fitness Function and Chromosome Locations



Example Problem & Solution

Step 3: The fitness ratio determines the chromosome's chance of being selected for mating.

- X5 & X6 have fair chance; X3 & X4 have less chance.

Chromosome Label	Chromosome String	Decoded Integer	Chromosome Fitness	Fitness Ratio
X1	1100	12	36	16.51
X2	0100	4	44	20.18
X3	0001	1	14	**6.42**
X4	1110	14	14	**6.42**
X5	0111	7	56	**25.68**
X6	1001	9	54	**24.77**

36+44+14+14+56+54 = 218

36/218 = 0.1651 = 16.51

Fitness Function in Roulette Wheel Distribution

1100 (12)=36

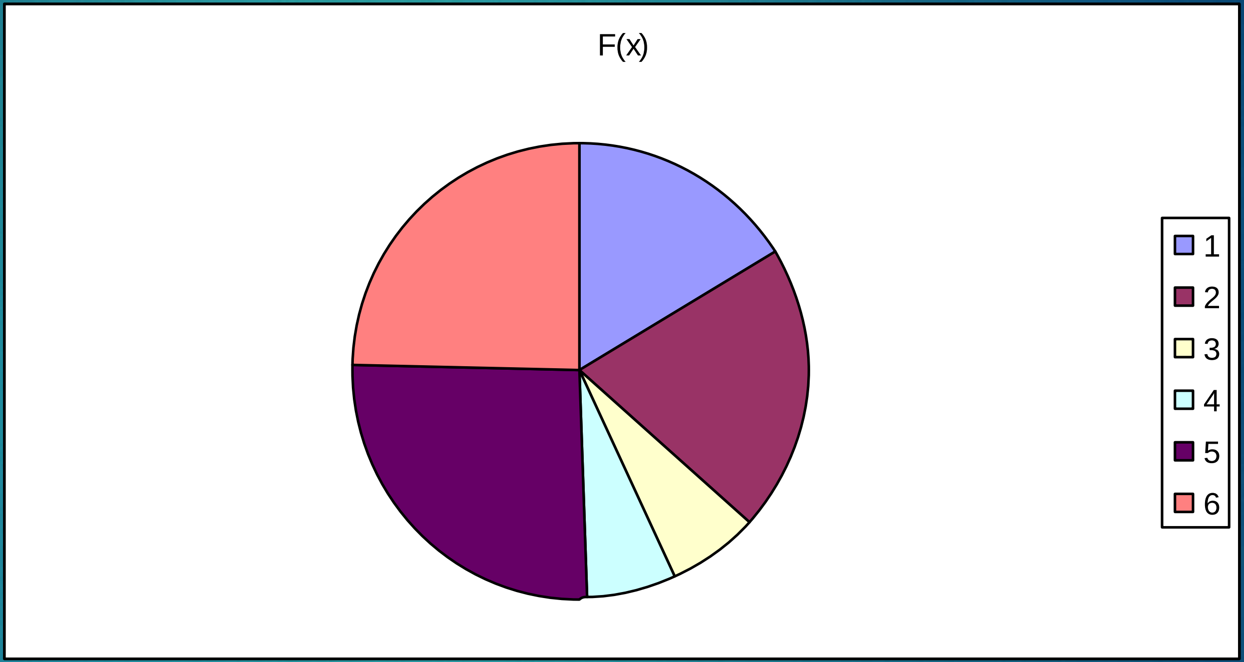
0100 (4)=44

0001 (1)=14

1110 (14)=14

0111 (7)=56

1001 (9)=54



Example Problem & Solution (Mating)

Step 4: Select chromosome for mating.

- First X6 & X2 become parents, 2nd X1 & X5, then X2 & X5.

Generation i		Crossover	Mutation
Chromosome Label	Chromosome String		
$X1_i$	1100	$X6_i$ 10-***01*** $X2_i$ 01-***00***	
$X2_i$	0100		
$X3_i$	0001	$X1_i$ 11-***00*** $X5_i$ 01-***11***	
$X4_i$	1110		
$X5_i$	0111	$X2_i$ 01-***00*** $X5_i$ 01-***11***	
$X6_i$	1001		

Example Problem & Solution (Crossover)

Step 5: Crossover Operator randomly chooses a crossover point where two parent chromosomes 'Break' and then exchange the chromosomes after that point.

Generation i		Crossover	Mutation
Chromosome Label	Chromosome String		
$X1_i$	1100	$X6_i$ 10-***01*** $X2_i$ 01-***00***	
$X2_i$	0100		
$X3_i$	0001	$X1_i$ 11-***00*** $X5_i$ 01-***11***	
$X4_i$	1110		
$X5_i$	0111	$X2_i$ 01-***00*** $X5_i$ 01-***11***	
$X6_i$	1001		

Example Problem & Solution (Crossover)

Step 5: If a pair of chromosomes does not cross over then chromosome cloning take places and the offspring are exact copies of each parent.

Generation i		Crossover	Mutation
Chromosome Label	Chromosome String		
$X1_i$	1100	$X6_i$ 10-***01*** $X2_i$ 01-***00***	$X6`_i$ 10***00***
$X2_i$	0100		$X2`_i$ 01**01**
$X3_i$	0001	$X1_i$ 11-***00*** $X5_i$ 01-***11***	$X1`_i$ 1111
$X4_i$	1110		$X5`_i$ 01**00**
$X5_i$	0111	$X2_i$ 01-***00*** $X5_i$ 01-***11***	$X2`_i$ 0100
$X6_i$	1001		$X5`_i$ 0111

Example Problem & Solution (Mutation)

Step 6: The mutation operator flips a randomly selected gene in a chromosome.

Generation i		Crossover	Mutation
Chromosome Label	Chromosome String		
$X1_i$	1100	$X6_i$ 10-*01* $X2_i$ 01-*00*	$X6`_i$ 10*00*
$X2_i$	0100		$X2`_i$ 01**01**
$X3_i$	0001	$X1_i$ 11-*00* $X5_i$ 01-*11*	$X1`_i$ 1111, $X1``_i$ 1011
$X4_i$	1110		$X5`_i$ 01**00**
$X5_i$	0111	$X2_i$ 01-*00* $X5_i$ 01-*11*	$X2`_i$ 0100, $X2``_i$ 0110
$X6_i$	1001		$X5`_i$ 0111

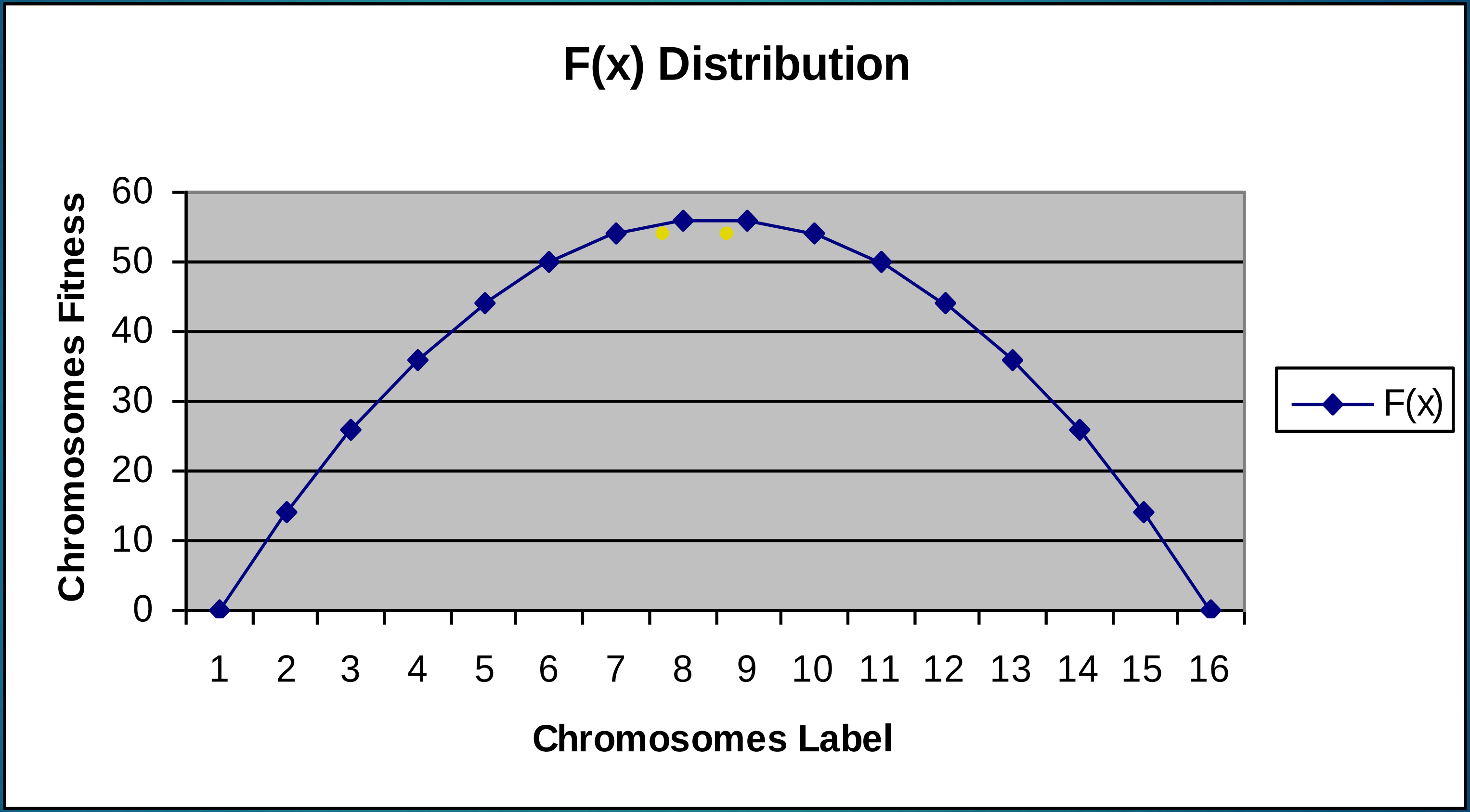
Example Problem & Solution

Step 7: The fitness ratio determines the chromosome's chance of being selected for mating.

- X5 & X6 have fair chance; X3 & X4 have less chance.

Generation i		Generation i+1		
Chromosome Label	Chromosome String	Chromosome Label	Chromosome String	Fitness
$X1_i$	1100	$X1_{i+1}$	1000	**56**
$X2_i$	0100	$X2_{i+1}$	0101	50
$X3_i$	0001	$X3_{i+1}$	1011	44
$X4_i$	1110	$X4_{i+1}$	0100	44
$X5_i$	0111	$X5_{i+1}$	0110	54
$X6_i$	1001	$X6_{i+1}$	0111	**56**

Fitness Function and Chromosome Locations



Applications of Genetic Algorithm

- Links:
- https://www.quora.com/What-are-some-real-world-applications-of-genetic-algorithms
- http://brainz.org/15-real-world-applications-genetic-algorithms/
- https://en.wikipedia.org/wiki/List_of_genetic_algorithm_applications

Recommended Textbooks

- [Negnevitsky, 2001] M. Negnevitsky " Artificial Intelligence: A guide to Intelligent Systems", Pearson Education Limited, England, 2002.
- [Russell, 2003] S. Russell and P. Norvig Artificial Intelligence: A Modern Approach Prentice Hall, 2003, Second Edition
- [Patterson, 1990] D. W. Patterson, "Introduction to Artificial Intelligence and Expert Systems", Prentice-Hall Inc., Englewood Cliffs, N.J, USA, 1990.
- [Lindsay, 1997] P. H. Lindsay and D. A. Norman, Human Information Processing: An Introduction to Psychology, Academic Press, 1977.

Thank You ☺